

Accessibility audit: OpenMRS O3 Reference Application

3.7.1

Standard: WCAG 2.2, Level AA · **Method:** WCAG-EM (sample-based) · **Date:** 7 October 2026 **Auditor:** Tayguara Dias Reis — Tech Lead, software quality (ISTQB CTFL)

Independent assessment of open-source software, performed as a portfolio piece. It is not issued, reviewed or endorsed by the OpenMRS project. Findings describe the tested version and sample only; they are not a legal opinion or a certification.

Executive summary

OpenMRS is an open-source electronic medical record system used by clinics and hospitals. In the US, software like it falls under HHS Section 504 (compliance date 11 May 2027 for recipients with 15+ employees), which requires WCAG 2.1 Level AA.

Result: the sampled screens do not conform to WCAG 2.2 AA. The audit found **18 failures** (10 from automated scanning, 8 from manual testing) across **11 success criteria**, 5 of them at Level A.

The most important pattern is about **keyboard focus**. Almost every function works with a keyboard, but in four places the user cannot see where the focus is, or the focus is lost or hidden:

- changing the page of a chart table throws the focus back to the top of the document (M1);
- the date-of-birth calendar can be operated with the arrow keys, but the selected day has no visible focus, so keyboard users can pick the wrong date without noticing (M5);
- at 200% zoom, a fixed bottom toolbar covers the focused controls (M6);
- a decorative container in the vitals card takes a Tab stop, does nothing and shows no focus (M2).

Two other findings have high impact for screen reader users: the patient search does not announce how many results it found (M3), and every screen has the same page title, "OpenMRS" (M8). At 320 px wide, the patient registration form cannot be used (M7).

Good news for remediation: two failures come from global components and appear on every screen (A2, and A1, which is already fixed in the current source), and most focus problems share one root cause. A consistent `:focus-visible` style, `scroll-padding` for the fixed bars and a handful of one-line attribute fixes would resolve about half of the findings.

1. Scope and method

Item	Value
System	OpenMRS O3 Reference Application 3.7.1 (official Docker images, pinned)
Environment	Local instance with the distribution's demo data
Standard	WCAG 2.2 Level AA (includes the WCAG 2.1 AA baseline required by ADA Title II and HHS 504)
Browsers	Chromium (Playwright 1.63), Firefox and Chrome on Linux, Chrome on Android
Assistive technology	Orca + Firefox (Linux), TalkBack + Chrome (Galaxy S22, Android)
Automated engine	axe-core 4.13.0 with the <code>wcag2a</code> , <code>wcag2aa</code> , <code>wcag21a</code> , <code>wcag21aa</code> and <code>wcag22aa</code> rule sets
Dates	7 October 2026

Why a local instance: the public demo is behind Cloudflare bot protection, which blocks automated scanning. A local, pinned instance also makes the audit reproducible.

Sample (WCAG-EM steps 2 and 3)

ID	Screen	Why it is in the sample
P1	Login	Entry point; WCAG 2.2 criterion 3.3.8
P2	Location picker	Mandatory step after login
P3	Home (service queues)	Most used screen; tables and filters
P4	Patient search and results	Dynamic results, status messages
P5	Patient registration	Long form: labels, errors, date input
P6	Patient chart — summary	Dense layout, paginated tables, fixed header
P7	Patient chart — vitals form	Side panel (workspace), focus on open/close
P8	Appointments	Calendar and filters; target size

Two **complete processes** were tested end to end, because WCAG requires a process to conform from start to finish:

- **Process A:** log in → choose location → register a patient (P1, P2, P5).
- **Process B:** search a patient → open the chart → record vitals (P4, P6, P7).

Severity

Severity	Meaning
High	Blocks a task or makes errors likely for some users
Medium	Task is possible but significantly harder, or information is lost
Low	Friction or non-conformance with limited impact

2. Results by success criterion

Criterion	Level	Result	Findings
1.3.1 Info and Relationships	A	Fails	A2
1.4.1 Use of Color	A	Fails	A5
1.4.3 Contrast (Minimum)	AA	Fails	A4
1.4.10 Reflow	AA	Fails	M7
2.4.2 Page Titled	A	Fails	M8
2.4.3 Focus Order	A	Fails	M1
2.4.7 Focus Visible	AA	Fails	M2, M5
2.4.11 Focus Not Obscured (Minimum)	AA	Fails	M6
2.5.8 Target Size (Minimum)	AA	Fails	A7, A8
4.1.2 Name, Role, Value	A	Fails	A1, A3, A6, A9, A10, M2, M4
4.1.3 Status Messages	AA	Fails	M3

Criteria verified without failures in the sample are listed in section 4. The full, criterion-by-criterion view is in the [Accessibility Conformance Report](#).

3. Findings

3.1 Manual testing

M1 — Focus is lost when changing the page of a chart table

2.4.3 Focus Order (A) · P6, P7 · Severity: Medium

On the chart widgets (Vitals, Biometrics, Conditions), changing the page with the page selector (arrow keys, Space and Enter) re-renders the table and the keyboard focus falls back to `<body>`. Keyboard and screen reader users lose their place on a long page.

Evidence: `document.activeElement` is the page `<select>` before the change and `<body>` after it. The select's `id` changes (`:r1b5:` → `:r19v:`), which shows the element is re-created, not updated.

The screenshot shows a web application interface for a patient's medical data. The left sidebar contains navigation links for Patient summary, Vitals & Biometrics, Medications, Orders, Results, Vitals, Allergies, Conditions, Immunizations, Procedures, Attachments, Programs, Appointments, and Billing history. The main content area is divided into sections for Vitals and biometrics, Biometrics, Conditions, Active Medications, Upcoming Medications, and Past Medications. The Vitals section displays a table with columns for Date and time, Temp (DEG C), BP (mmHg), Pulse (beats/min), R. Rate (breaths/min), and SpO2 (%). The Biometrics section displays a table with columns for Date and time, Weight (kg), Height (cm), BMI (kg / m²), and MUAC (cm). The Conditions section displays a table with columns for Condition, Date of onset, and Status. The Active Medications section displays a message: "There are no active medications to display for this patient". The Upcoming Medications section displays a message: "There are no upcoming medications to display for this patient". The Past Medications section displays a message: "There are no past medications to display for this patient". On the right side, the developer console is open, showing a JavaScript error: "Uncaught SyntaxError: Unexpected token. Identifier 'class' must be followed by a colon." The error message is: "Uncaught SyntaxError: Unexpected token. Identifier 'class' must be followed by a colon." The error is located in the file `patient-chart-pagination_pagination_v222.js` at line 17, column 17. The error message is: "Uncaught SyntaxError: Unexpected token. Identifier 'class' must be followed by a colon." The error is located in the file `patient-chart-pagination_pagination_v222.js` at line 17, column 17. The error message is: "Uncaught SyntaxError: Unexpected token. Identifier 'class' must be followed by a colon." The error is located in the file `patient-chart-pagination_pagination_v222.js` at line 17, column 17.

Fix: keep the same pagination instance across renders (stable key), or move focus back to the selector after the page changes.

M2 — A container in the vitals chart takes focus, does nothing and shows no focus

2.4.7 Focus Visible (AA) and 4.1.2 Name, Role, Value (A) · P6, P7 · Severity: Medium

In the sequence Show more → Sticky note → next Tab, focus lands on the vitals chart header before reaching "Record vitals". The header is a `div` with `role="button"` and `tabindex="0"`: it is announced as a button, has no visible focus, Enter does nothing, Space scrolls the page, and a mouse click does nothing either. It is a dead Tab stop. Neighbouring controls ("Sticky note") show a clear focus outline. This is the same element as automated finding A6 (nested interactive controls).

M4 — A button nested inside a link in search results

4.1.2 Name, Role, Value (A) · P4 (global search panel) · Severity: Medium

Each search result is a link to the patient chart (`<a href= ".../chart/">`) that contains a button: the "Active Visit" tag, which opens a toggletip. Interactive content inside a link is invalid HTML, and screen readers handle it inconsistently (the button may be hidden, or the link name may include the button text).

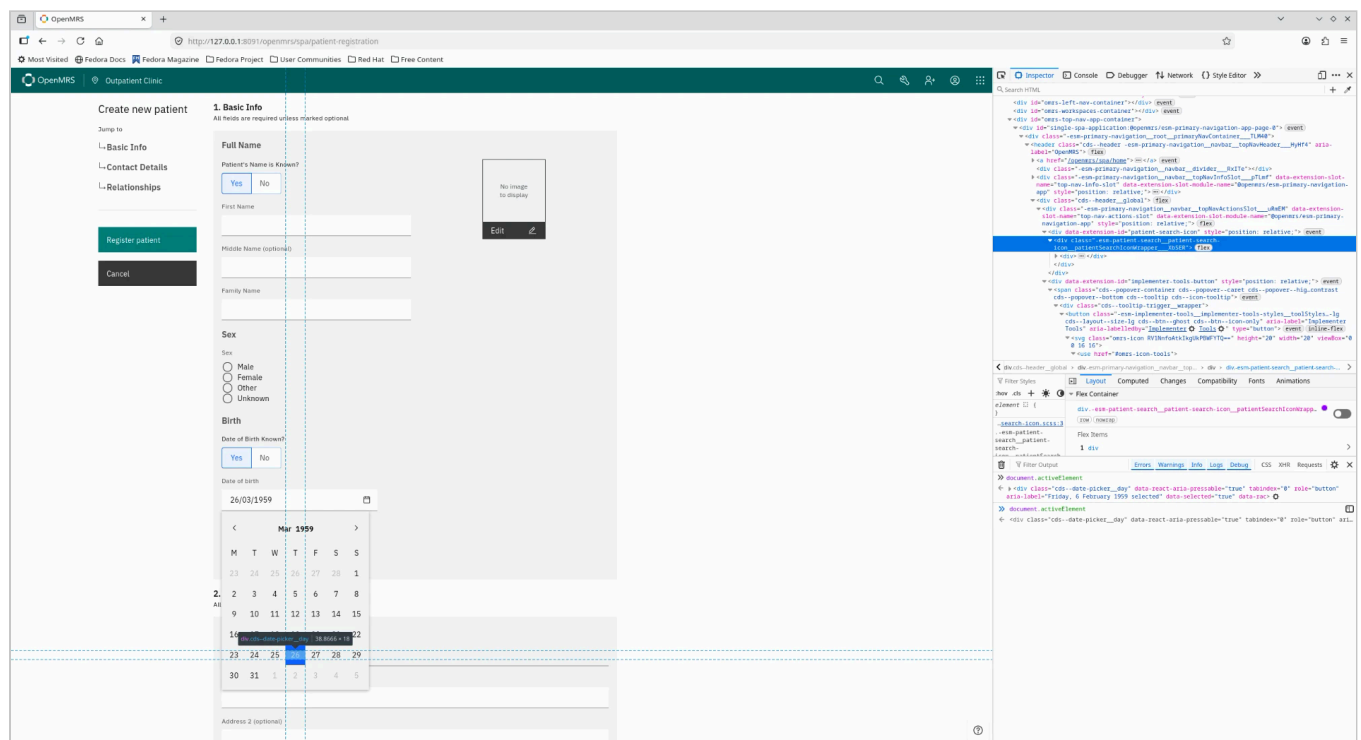
Fix: move the tag out of the link, or render it as a non-interactive label inside search results.

M5 — Calendar days have no visible focus

2.4.7 Focus Visible (AA) · P5 (date of birth) · Severity: High

Typing the date in the dd/mm/yyyy segments works. In the calendar pop-up, Tab moves through "<", the year and ">", and the next Tab moves focus into the grid of days, but the focused day has no visible indicator. The arrow keys move the selection and change the value of the field, and Enter confirms, all without visual feedback. During testing the grid seemed not to respond; in fact it was being navigated blindly, and the date changed from 06/02/1959 to 26/03/1959 without the user seeing why. The calendar trigger button also receives focus without an indicator.

Evidence: after the Tab, `document.activeElement` is `<div class="cds--date-picker__day" role="button" tabindex="0" aria-label="Friday, 6 February 1959 selected">`; after arrow keys and Enter it is "Thursday, 26 March 1959 selected".



Fix: add a `:focus-visible` (or `[data-focus-visible]`) style to calendar days and to the trigger button, with a 2 px outline and at least 3:1 contrast. In a medical record, a silently wrong date of birth is a data-quality risk, which is why this is rated High.

M6 — At 200% zoom, the fixed bottom toolbar hides the focused control

2.4.11 Focus Not Obscured (Minimum) (AA) · P6 at 200% zoom · Severity: Medium

At 200% zoom the chart switches to its tablet layout: the left menu becomes a hamburger menu and the chart actions become a toolbar fixed to the bottom of the screen. Moving forward with Tab, the controls of the Vitals card (table view toggle, "Add", and the following ones) receive focus behind that toolbar: first partly, then completely hidden. The page does not scroll, because for the browser the element is already inside the viewport. Without zoom the same path passes, because the bottom toolbar does not exist in the desktop layout. Zoom users often have low vision, the group that depends most on seeing the focus.

OpenMRS | Outpatient Clinic

Joshua Johnson ♂ Male Active Visit Actions

5 yrs · 25-Sept-2021 · OpenMRS ID: **100008E** Show more

Sticky note

Vitals and biometrics Today, 08:16 AM Vitals history Record vitals

BP	Heart rate	R. rate	SpO2
101 / 62 mmHg	81 beats/min	20 breaths/min	97.9 %
Temp	Weight	Height	BMI
36.9 DEG C	90 kg	191 cm	24.7 kg / m ²

Table view

Order basket Visit note Task list Clinical forms Patient lists

OpenMRS | Outpatient Clinic

Joshua Johnson ♂ Male Active Visit Actions

5 yrs · 25-Sept-2021 · OpenMRS ID: **100008E** Show more

Sticky note

Vitals and biometrics Today, 08:16 AM Vitals history Record vitals

BP	Heart rate	R. rate	SpO2
101 / 62 mmHg	81 beats/min	20 breaths/min	97.9 %
Temp	Weight	Height	BMI
36.9 DEG C	90 kg	191 cm	24.7 kg / m ²

Vitals

Order basket Visit note Task list Clinical forms Patient lists

Fix: set `scroll-padding-bottom` (and `scroll-padding-top`) on the scroll container to the height of the fixed bars, so the browser scrolls focused elements clear of them.

M7 — Content is cut off at 320 px wide; registration cannot be completed

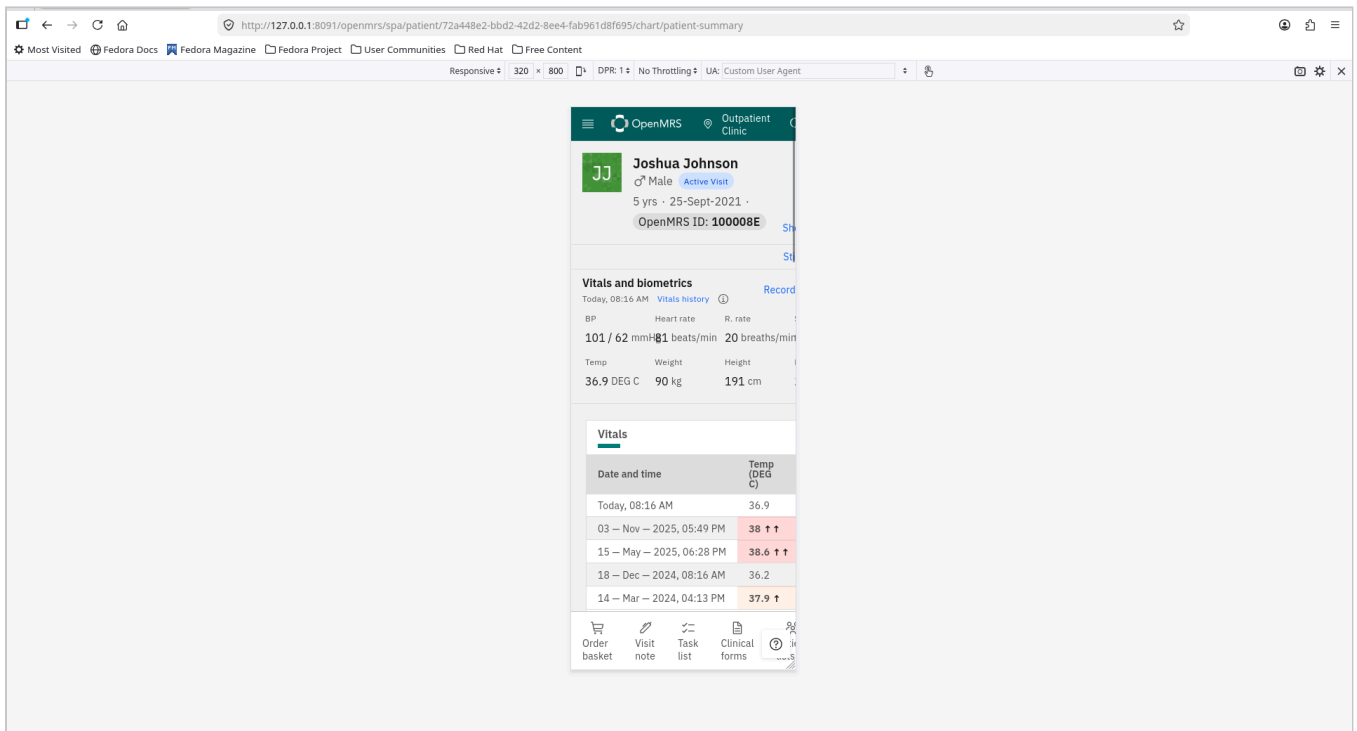
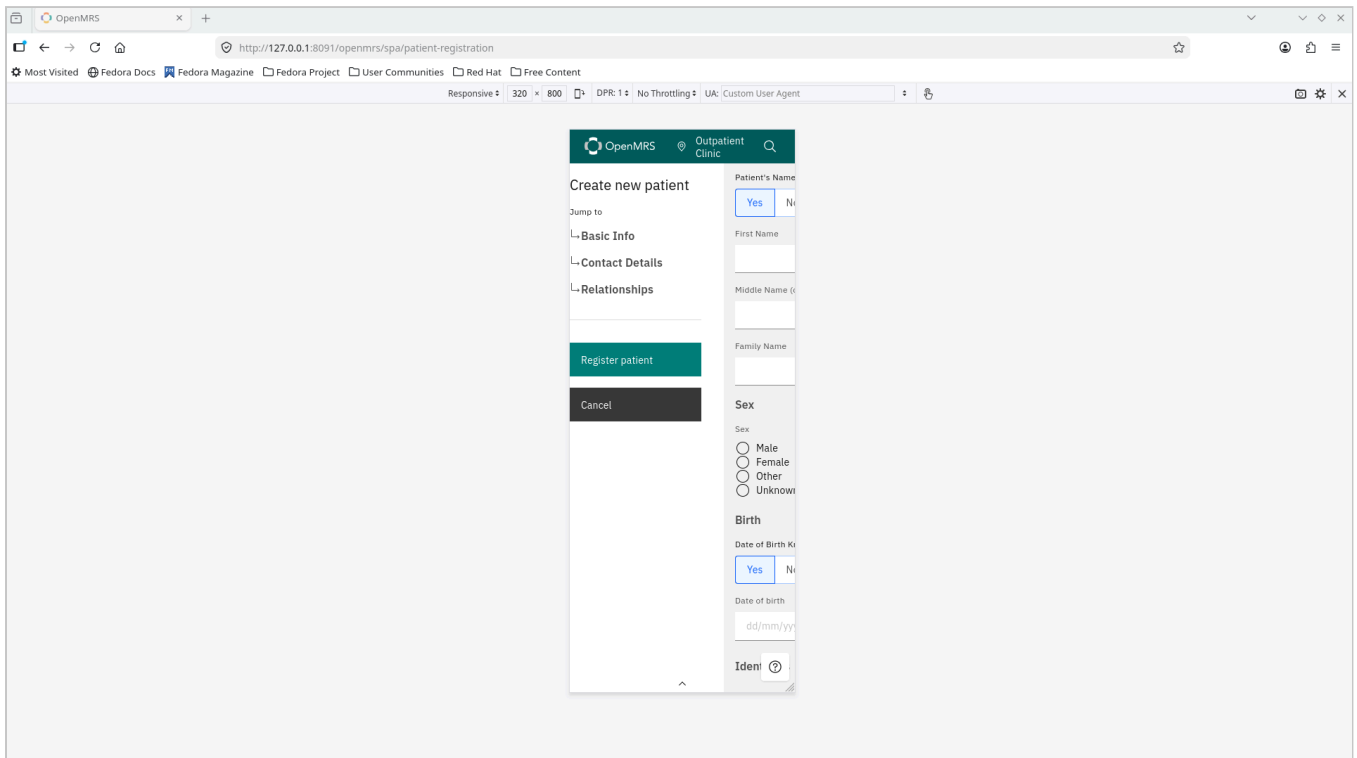
1.4.10 Reflow (AA) · P5, P6 at 320 px · Severity: High

At 320 CSS pixels (equivalent to 1280 px at 400% zoom) there is no horizontal scrollbar, but content that does not fit is **clipped** with no way to reach it:

- **Registration:** the two-column layout does not stack. The "Jump to" menu and the action buttons take two thirds of the width; the form is squeezed and cut (labels cut in half, the "No" button and the inputs cut, the date field unreadable, the help button over "Identifiers"). The form cannot be completed at this width.

- **Chart:** the top bar icons, "Show more", "Sticky note", "Record vitals", the right column of the vitals summary (SpO2, BMI) and "Patient lists" in the bottom toolbar are cut; in the summary, values overlap ("mmHg" over "81 beats/min").

Data tables with their own horizontal scroll would be an allowed exception; these are not tables.



Fix: stack the "Jump to" menu and buttons above the form on narrow screens; let the patient header and action rows wrap; make the vitals summary a single column below about 480 px; move overflow icons from the top bar into the menu; reserve space for the help button above the bottom toolbar.

M8 — Every screen has the same page title

2.4.2 Page Titled (A) · All screens (P1–P8) · Severity: Medium

All eight screens have `<title>OpenMRS</title>`. Screen reader users hear the title first when a page loads and use it to tell browser tabs apart; with identical titles they cannot tell the login, a patient chart or the registration form apart without exploring the page.

Evidence: the `title` recorded for each screen in the automated scan results.

Fix: set the document title per route, for example "Joshua Johnson — Patient chart — OpenMRS" or "Register patient — OpenMRS".

3.2 Automated scanning (axe-core 4.13.0)

Screen	Violations (rules)	Items needing manual review
P1 Login	0	1
P2 Location picker	1	0
P3 Home (service queues)	3	15
P4 Patient search	4	14
P5 Patient registration	4	15
P6 Patient chart — summary	6	54
P7 Patient chart — vitals form	5	52
P8 Appointments	5	14

The violations are not independent: two come from global components and repeat on almost every screen.

ID	Criterion	Level	Where	What happens	Suggested fix	Severity
A1	4.1.2 Name, Role, Value	A	Global (P2–P8)	The help button (bottom right) has no accessible name; screen readers announce only "button"	<code>aria-label="Help"</code> on the help menu button	Medium
A2	1.3.1 Info and Relationships	A	Global (P3–P8)	User menu: the <code></code> has <code><div></code> children, and the <code></code> items end up outside a valid list	Render each extension inside an <code></code> , or remove the wrapping <code><div></code>	Low
A3	4.1.2 Name, Role, Value	A	P4	The refine-search form uses <code>role="refine-search"</code> , which is not an ARIA role	Remove the role (or use <code>role="search"</code>)	Low
A4	1.4.3 Contrast (Minimum)	AA	P5	The dd/mm/yyyy placeholders of the date of birth have 1.72:1 contrast (<code>#c5c5c5</code> on white; 4.5:1 required)	Darken the date picker placeholder token	Low
A5	1.4.1 Use of Color	A	P6	The "See all" links differ from the surrounding text only by colour, at 1.55:1 between them (3:1 required)	Underline the links or raise the contrast	Low
A6	4.1.2 Name, Role, Value	A	P6, P7	The vitals card header is a <code>div</code> <code>role="button"</code> that contains a link and buttons (nested interactive controls); see M2	See M2	Medium
A7	2.5.8 Target Size (Minimum)	AA	P6, P7	"Vitals history" is 16 px tall; "View normal ranges" is 16×16 px (24×24 px or equivalent spacing required)	Enlarge the clickable area with padding	Low
A8	2.5.8 Target Size (Minimum)	AA	P8	The day and month segments of the date picker are 16×18 px	Same padding fix in the date component	Low

ID	Criterion	Level	Where	What happens	Suggested fix	Severity
A9	4.1.2 Name, Role, Value	A	P8	Two filter dropdowns have <code>aria-labelledby</code> pointing to empty labels	Fill the label or add <code>aria-label</code>	Medium
A10	4.1.2 Name, Role, Value	A	P8	The batch-actions bar has <code>aria-hidden="true"</code> but contains focusable elements: keyboard users reach controls that screen readers do not announce	Use <code>inert</code> (or <code>hidden</code>) while the bar is hidden	Medium

Items the tool could not decide (counted across all screens, to be reviewed manually in a full audit): text contrast over coloured or gradient backgrounds (76, mostly the alert colours in the vitals tables), ARIA attributes that may be prohibited or invalid (87, mostly from the same components), one table header without data cells (the empty service queue), and the bypass-blocks check on the login page.

3.3 Status on the current source code

Each finding was checked against the `main` branch of the three OpenMRS frontend repositories on 7 October 2026 (details, permalinks and code quotes in [source-verification.md](#)):

Status on main	Findings
Fixed and released	A1 (help button now has <code>aria-label="Help menu"</code> , esm-core v11.0.0, 6 Oct 2026)
Fixed, not yet released	A3 (invalid role replaced; fix merged after the latest patient-management release)
Still present	A2, A5, A6/M2, M3, M4, M5 (calendar days), M6, M8
Needs a runtime re-test	M1 (the source shows no forced re-mount of the pagination; the focus loss may come from the Carbon component)

The source review also explains M2: the vitals header has an `onClick` that toggles a state no other code reads, so the "button" has no effect even with a mouse. Two further observations from the source, not tested at runtime: each patient card in the full-page search uses `role="banner"` (likely several banner landmarks on one page), and the search loading skeletons use `role="progressbar"` without an accessible name.

4. Verified without failures

Check	Criterion	Result
Tab and Shift+Tab through the chart without zoom: the focused element stays visible behind the fixed top bar and help button	2.4.11	Passes
Zoom to 200% on registration, search and chart: content reflows into one column, nothing cut or overlapping	1.4.4	Passes
Submitting the empty registration form: submission is blocked, focus moves to the first empty required field, errors are given in text next to each field	3.3.1	Passes
Error text is programmatically tied to the field: <code>aria-invalid="true"</code> and <code>aria-describedby="familyName-error-msg"</code>	3.3.1, 4.1.2	Passes
Calendar pop-up: Esc closes it and Tab leaves it	2.1.2	Passes
Typing in the search field with TalkBack: letters are entered, focus stays in the field, no unexpected context change	3.2.2	Passes
The help button is in the same place on every screen	3.2.6	Passes

OpenMRS Outpatient Clinic

Create new patient

Jump to

- Basic Info
- Contact Details
- Relationships

Register patient

Cancel

1. Basic Info

All fields are required unless marked optional

Full Name

Patient's Name is Known?

Yes No

First Name

Rubens

Middle Name (optional)

Family Name

Family name is required

Sex

Sex

Male

Female

Other

Unknown

Gender is required

Birth

Date of Birth Known?

Yes No

Date of birth

dd/mm/yyyy

Birthday is required

Identifiers

Configure →

Recommendations that are not failures:

- The yellow error toast disappears on its own. The same errors stay next to each field, so no information is lost (2.2.1 does not apply), but the toast is the only summary of all errors; keep it until the user dismisses it.
- The same field has different names in its label, its error and the toast: "Date of birth" / "Birthday is required" / "birthdate", and "Sex" / "Gender is required". Use the visible label in error messages.
- On repeated submits, two "Please fill out this field." bubbles overlap and the toasts pile up.

Not an OpenMRS failure: with Orca and Firefox, typing in the global search stopped after the first letter. The same test passed with TalkBack, so it was recorded as an Orca limitation.

5. Remediation roadmap

Priority	Action	Fixes	Effort
1	Consistent :focus-visible style in the design system, including calendar days and icon buttons	M2, M5	Small
2	scroll-padding on the scroll container for the fixed top and bottom bars	M6	Small
3	role="status" around the search results count	M3	Small
4	Per-route document titles	M8	Small
5	One-line attribute fixes: help button name, invalid role, empty labels, inert for hidden bars	A1, A3, A9, A10	Small
6	Stable pagination instance, or focus restored after page change	M1	Small-medium
7	Remove the fake button around the vitals header; take the tag out of search result links	M2/A6, M4	Medium
8	Responsive fixes for 320 px: registration layout first, then chart header and vitals summary	M7	Medium
9	Contrast and target size adjustments in the design tokens	A4, A5, A7, A8	Small

After remediation, an automated accessibility check in CI (for example axe-core with Playwright on the main flows) keeps these fixes from regressing release after release.

6. Limitations

- **Sample, not full coverage.** Eight screens and two processes. Other modules (orders, laboratory, billing, wards, forms) were not tested.
- **Assistive technology coverage.** NVDA and JAWS (Windows) and VoiceOver (iOS, macOS), the most used screen readers in the US, were not tested in this round. Semantic findings (M3, M4, M8) were confirmed from the code; screen reader behaviour is limited to Orca and TalkBack.
- **Version.** Findings refer to 3.7.1. Section 3.3 gives their status on the current source code; A1 and A3 are already fixed there.
- **Demo data.** One demo visit was created through the REST API to reach the vitals form.

7. How to reproduce

```
cd audits/2026-10-openmrs-o3/scan
docker compose up -d      # OpenMRS 3.7.1 on http://127.0.0.1:8091/openmrs/spa (first start: up to ~1 h)

cd ../../../../tools/axe-scan
npm ci
npm run login -- http://127.0.0.1:8091/openmrs/spa/login # log in, pick a location, close the window
node scan.mjs ../../audits/2026-10-openmrs-o3/scan/pages.json --auth .auth/state.json
```

The login uses the distribution's default demo credentials from the OpenMRS documentation. The session is stored in `.auth/` (ignored by git) and expires after about 30 minutes of inactivity. The P7 page needs an active visit for the patient in `pages.json`.